

Project Planning

AI Learning Assistant

Project Title

AI Learning Assistant

Domain

Artificial Intelligence | Education Technology

1. Introduction

Project planning is an important phase in software development that defines the project's objectives, timeline, tasks, and resources. It helps ensure that the project is completed efficiently within the available time.

The AI Learning Assistant project was planned by dividing the work into multiple phases, including requirement analysis, design, development, testing, and deployment.

2. Project Objectives

The main objectives of the project are:

- Develop an AI-powered learning assistant.
 - Detect students' emotions from text input.
 - Generate personalized study guidance using Google Gemini AI.
 - Store interaction history.
 - Display learning analytics through a dashboard.
 - Provide a simple and user-friendly interface.
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3. Project Timeline

Phase	Activity	Duration
Phase 1	Requirement Analysis	1 Week
Phase 2	System Design	1 Week
Phase 3	Development	2 Weeks
Phase 4	Testing	1 Week
Phase 5	Documentation	1 Week
Phase 6	Final Deployment	2 Days

4. Work Breakdown Structure (WBS)

Task 1

Project Idea Selection

Task 2

Requirement Gathering

Task 3

System Design

Task 4

Development

- User Interface

- Emotion Detection Module
- Gemini AI Integration
- Logger Module
- Dashboard Module

Task 5

Testing

Task 6

Documentation

Task 7

Project Demonstration

5. Resource Planning

Hardware

- Laptop/Desktop
- Internet Connection

Software

- Python
 - Streamlit
 - Visual Studio Code
 - Google Gemini API
 - Git
 - GitHub
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6. Risk Management

Risk

Solution

Internet Failure Retry after connection is restored

Gemini API Issues Handle errors gracefully

Incorrect User Input Validate input before processing

Data Loss Maintain backup of project files

7. Milestones

-  Problem Identification
 -  Requirement Analysis
 -  Project Design
 -  Development
 -  Testing
 -  Documentation
 -  Final Demonstration
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8. Expected Deliverables

The project will deliver:

- AI Learning Assistant Application
 - Source Code
 - Documentation
 - Dashboard
 - GitHub Repository
 - Project Demonstration
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9. Conclusion

The Project Planning phase provided a structured roadmap for developing the AI Learning Assistant. By dividing the work into manageable phases and identifying resources, timelines, and risks, the project was completed systematically and efficiently.